A graph can tell the story of motion

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Average velocity / ES: velocidad media / PT: velocidade média

$\bar{v} = \Delta x \div \Delta t$. Carries direction; summarizes a whole interval.

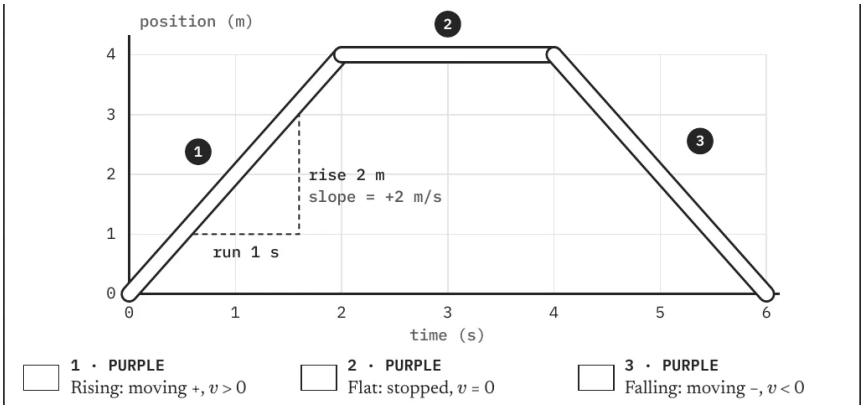
Also on your note page: Average speed, Slope, Instantaneous velocity.

B READ THE RELATIONSHIP

Average velocity = displacement ÷ elapsed time

As you read, find: rise · run · slope.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A straight x – t line passes through (2 s, 14 m) and (6 s, 6 m). Find the velocity, then predict x at 8 s.

Given (2 s, 14 m) (6 s, 6 m) predict at 8 s

Rule $v = \Delta x \div \Delta t$ $x = x_i + v\,t$

Work $v = (6 - 14) \div (6 - 2)$ x $(8) = 6 + (-2)(2)$

Answer –2 m/s; 2 m if the line continues

E SAVE FOR THE FINAL MEETING

Average velocity, Δx ÷ Δt from two tracked positions, is the desk's first estimate of how fast XJ closes in. The slope of its x–t graph shows whether that rate holds.

Read the physics: Average velocity

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Find the start and finish on a position–time graph. Read position vertically and time horizontally. A rising line means position increases. A flat line means the object stays in one place.

02 HOW IT WORKS

Divide change in position by elapsed time. The slope is average velocity over that interval. A falling line has negative slope: the object moves toward smaller positions.

03 WHAT TO CHECK

Speed uses total distance. A return trip can have zero average velocity even when the object moved.

04 TRY THE EXAMPLE FROM YOUR NOTES

A straight x – t line passes through (2 s, 14 m) and (6 s, 6 m). Find the velocity, then predict x at 8 s. –2 m/s; 2 m if the line continues.

Model check: Average velocity = displacement ÷ elapsed time.

05 CONNECT TO XJ

Average velocity, $\Delta x \div \Delta t$ from two tracked positions, is the desk's first estimate of how fast XJ closes in. The slope of its x–t graph shows whether that rate holds.

SAY IT BEFORE YOU WRITE

What does the sign of the slope tell you? The line slopes ___. The object moves ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Average velocity

ES: velocidad media / PT: velocidade vetorial média

Meaning in my words	How can you tell?
How would you explain average velocity to a friend? In my words, this means ___.	A cart travels out and back. Which two points determine its average velocity? I can tell because ___.
Example or sketch	**Non-example and why**
Right is positive. A cart goes from +8 m to +2 m in 3 s. What is its average velocity? My example shows ___.	"Average the two speeds, and you always get average velocity." What is missing? I would change ___ because ___.

Average speed

ES: rapidez media / PT: rapidez média

Meaning in my words	How can you tell?
How would you explain average speed to a friend? In my words, this means ___.	On a trip out and back, which distance goes on top of the fraction? I can tell because ___.
Example or sketch	**Non-example and why**
You walk 3 m out and 3 m back in 4 s. What is your average speed? My example shows ___.	"I got back to the start, so my average speed was zero." What would you say? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Slope

ES: pendiente / PT: inclinação

Meaning in my words	How can you tell?
How would you explain slope to a friend? In my words, this means ___.	On a position–time graph, what does a line falling to the right tell you? I can tell because ___.
Example or sketch	**Non-example and why**
A line goes from 2 m at 0 s to 8 m at 3 s. What slope would you label? My example shows ___.	The graph reaches 8 m. Does that make the slope 8 m/s? I would change ___ because ___.

Instantaneous velocity

ES: velocidad instantánea / PT: velocidade instantânea

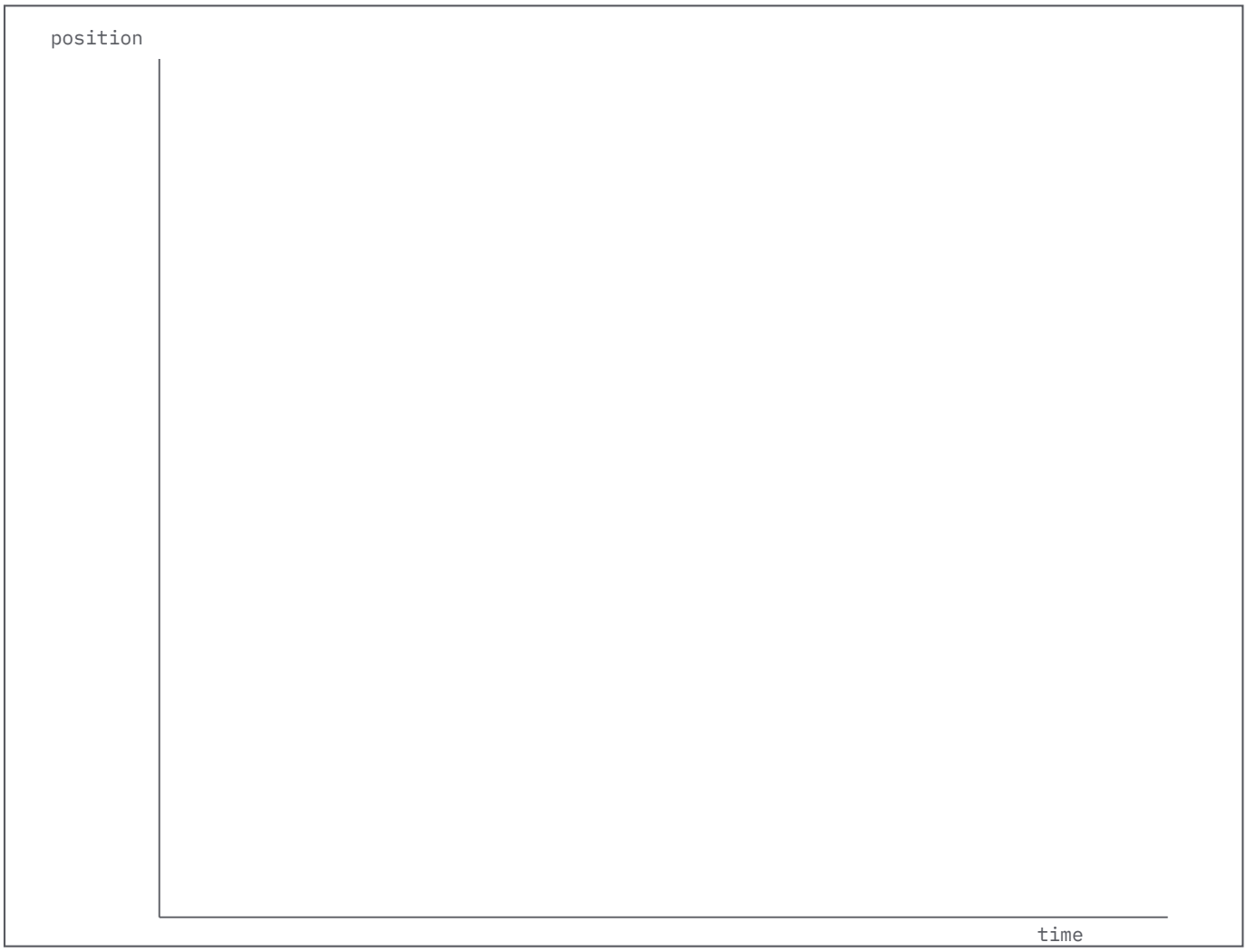
Meaning in my words	How can you tell?
How would you explain instantaneous velocity to a friend? In my words, this means ___.	On a curved position–time graph, what line would you draw at exactly 2 s? I can tell because ___.
Example or sketch	**Non-example and why**
What does a tangent slope of −3 m/s at 2 s tell the camera operator? My example shows ___.	A partner joins the first and last points of a curved trip. Which velocity does that slope give? I would change ___ because ___.

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Sketch a falling position–time line. Label the signed rise, positive run and velocity unit.

Labels you can use: rise · run · slope.



Tell a partner

The line slopes ___. The object moves ___.

Is it really slowing down?

03

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Use a position-time graph to find velocity.

CHOOSE ONE JOB TOGETHER

RADAR OPERATOR ☐

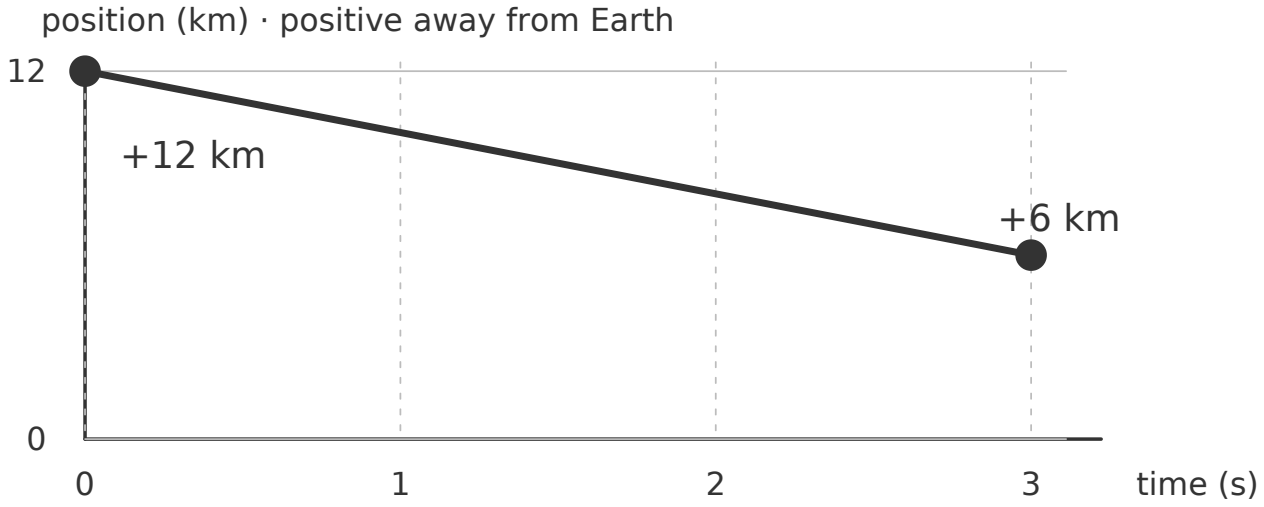
Calculate the signed average velocity. Mark the two sightings on the graph so the next operator can check your result.

TRAFFIC BROADCASTER ☐

Report a speed and direction. Tell viewers whether these two sightings show that XJ is slowing down.

Word help: position / velocity / toward

Radar sees XJ at **+12 km**, then **+6 km** three seconds later. Positive points away from Earth.



01 OUR FIRST CALL

What can you report about XJ's motion? Use the graph to support your call.

XJ moves ___ at an average of ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Check the new caption. Write the correction you would send or say on air.

Now we would say ___ because ___.

Switch speakers. What can these two sightings tell you about average velocity? What can they not tell you about motion between sightings?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks